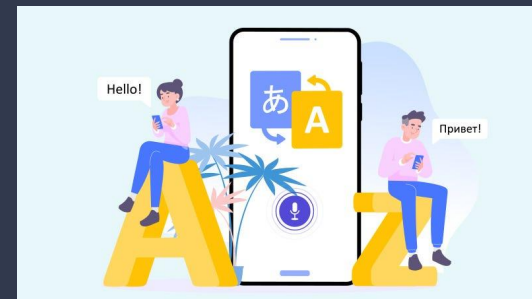


Phrase-Based & Neural Unsupervised Machine Translation



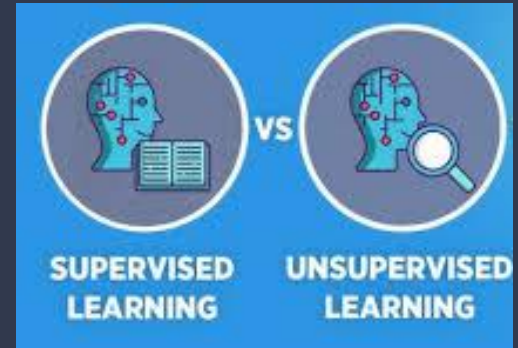
Prof. Laura Nenzi
Nooshin Behrooz
July 2026



- 1. Introduction: The Problem**
- 2. The Three key Principles**
 - 2.1 Initialization
 - 2.2 Language Modeling
 - 2.3 Iterative Back-Translation
- 3. The Two Models**
 - 3.1 Neural Machine Translation-NMT
 - 3.2 Phrase-Based Statistical Machine Translation - PBSMT
- 4. The Results and Comparison**
 - 4.1 Paper Results
 - 4.2 Our Implementation Results
- 5. Conclusion**

Table of Contents

1. *Introduction: The Problem*



1. Introduction: The Problem

Before publishing this paper (2018)

People believed: No parallel data → No translation system

Translation was considered almost **impossible** without bilingual sentence pairs.

Why? Because MT Systems, which are very effective,
but their success strongly **relies on large amounts of parallel sentences.**

1. Introduction: The Problem

1) Why does MT need parallel corpora?

MT learns by matching sentence pairs across languages to learn word, phrase, and sentence alignments.

Examples of Supervised MT systems:

Neural Machine Translation (NMT) (Encoder–Decoder models)

- Seq2Seq LSTM (2014)
- Transformer (2017)
- T5 (2019/2020)
- mBART (2020)
- NLLB (2022)

Statistical Machine Translation (SMT)

- Phrase-Based Statistical Machine Translation (PBSMT)
- Moses (popular PBSMT toolkit)

2) Why is this a problem?

Many language pairs lack sufficient parallel corpora, especially low-resource languages.

- French–Urdu
- Minority languages

Situation	Expected Performance
Lots of direct LA ↔ LB	Excellent
No Di LA ↔ LB Data, but lots of En links	Good
Very little data for both languages	Weak

1. Introduction: *The Solution*

The paper introduces:

Phrase-Based & Neural Unsupervised Machine Translation

Goal:

translating between two languages **without parallel sentence pairs**
using only **monolingual corpora** in each language

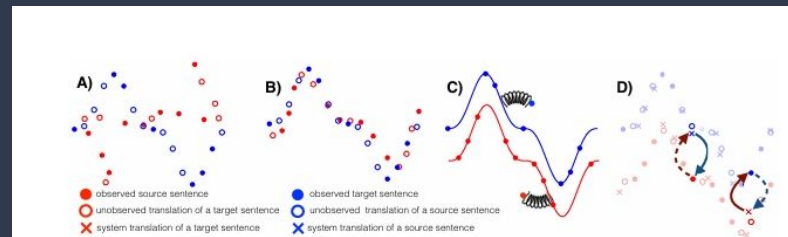
How?

2. The Three key Principles

2.1 Initialization

2.2 Language Modeling

2.3 Iterative Back-Translation



2. The Three key Principles

2.1 Initialization

Why is it needed?

There is **no parallel data**, so the model **needs a weak starting bridge** between the two languages

Main idea:

The system **starts** with an approximate **cross-lingual alignment**, so training does **not begin from pure randomness**.

In NMT:

[Shared Embeddings] from both monolingual corpora

In PBSMT:

An **Initial Phrase Table** is built from a bilingual dictionary inferred from cross-lingual embeddings

Why is it important?

Bad Initialization can send the system in the **Wrong Direction**, and **Back-Translation** may not fully fix it

2. The Three key Principles

2.2 Language Modeling

Why is it needed?

After initialization, **translations are still noisy and unnatural**

Main idea:

Train a language model on each language separately, so the system **learns what a correct sentence looks like**, the **Structure** and **Fluency of each language**, and acting as a correction mechanism for noisy outputs)

In NMT:

Use a **Denoising Autoencoder** (real sentence → add noise → reconstruct clean sentence)

In PBSMT:

Use an **n-gram language model** (such as KenLM, to score and prefer fluent translations)

2. The Three key Principles

2.3 Iterative Back-Translation

Why is it needed?

Even with initialization and language modeling, there are **still no parallel sentence pairs**

Main idea:

Automatically generate pseudo-parallel data by translating sentences between the two languages

How does it work?

1. Translate a sentence to the other language.
2. Translate it back.
3. Compare with the original sentence.
4. Update the model.

Why iterative?

- Early iterations → poor translations
- Later iterations → better synthetic data
- Final iterations → stronger translation models

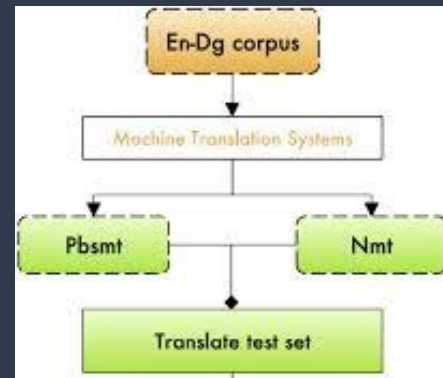
Key idea:

Back-translation turns an **unsupervised problem** into a **supervised learning problem** using automatically generated sentence pairs

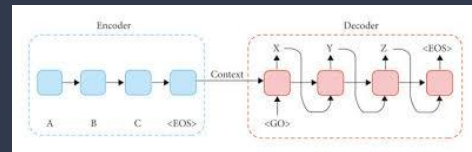
3. The Two Models

3.1 Neural Machine Translation-NMT

3.2 Phrase-Based Statistical Machine Translation-PBSMT



3. The Two Models: 3.1 Neural Machine Translation (NMT)



1. Initialization

Goal: Create a **Shared Embedding Space** for English and Italian **without parallel sentences**

Step 1: Tokenization English/Italian:

I love pizza $\rightarrow$ [I, love, pizza]; Italian: A Luca piace pizza $\rightarrow$ [A, Luca, piace, pizza]

Step 2: Train Monolingual Embedding Vectors

love $\rightarrow$ [0.1, 0.8], pizza $\rightarrow$ [0.2, 0.5]; amare $\rightarrow$ [0.3, 0.9], pizza $\rightarrow$ [0.7, 0.1]

Step 3: Align Embedding Spaces (Find rotation matrix W^*)

$W(\text{Emb_En}) \approx \text{Embedding_It}$: $W(\text{love}) \approx \text{amare}$

Cosine similarities: $\text{Cos}(W(\text{love}), \text{amo}) \dots \dots$ (IR)

love $\rightarrow$ amare 0.98

love $\rightarrow$ adorare 0.72

love $\rightarrow$ piacere 0.65

Build a temporary bilingual lexicon by Choosing the **Nearest-Neighbor Retrieval**

Step 4: Compute a better W

Find the rotation matrix that makes all these pairs as close as possible. (w_1 better than w_0)

Step 5: Repeat Step 3 and 4 to reach final W and Final Shared Bilingual Embeddings:

[Aligned English embeddings] and [Italian embeddings]: $W(\text{love_En}) \leftrightarrow \text{amare_It} : [0.31, 0.93] \leftrightarrow [0.3, 0.9]$

Train EN/IT embeddings

Find W

$\cos(W(\text{love}), \text{amo})$

$\cos(W(\text{home}), \text{casa})$

...

Temporary bilingual lexicon

love $\rightarrow$ amo

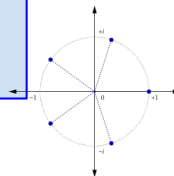
home $\rightarrow$ casa

Improve W

Repeat a few times

Final W

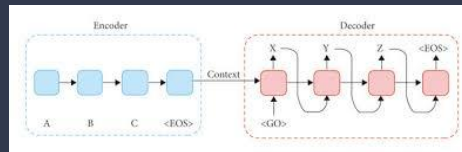
Final aligned embeddings



***Isomorphism Hypothesis:** The geometric structures of language spaces share similar topologies.

Calculation W , Gradient-based methods or SVD (Singular Value Decomposition)

3. The Two Models: 3.1 NMT



2. Language Modeling (Denoising Autoencoder)

Goal: Learn grammar, word order and fluency of each language separately using **monolingual corpora**.

ENCODER

INPUT: Corrupted Sentence_En: I l ve pizz

↓
Embedding → Self-Attention → Feed Forward
(initialized by **Shared Bilingual Embeddings**)

↓
OUTPUT: Hidden States ($B \times N \times H$)

DECODER

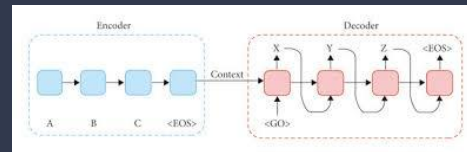
INPUT: Clean Sentence_En (Teacher Forcing): <START> I love pizza

↓
Decoder Hidden States → **Cross Attention** with **Encoder Hidden States**

↓
OUTPUT: Prediction Matrix ($B \times N \times V$), Ex: I like pizza

LOSS_LM_En: Prediction vs Original Clean Sentence
Repeat for It corpora

3. The Two Models: 3.1 NMT



3. Iterative Back-Translation

Goal: Create pseudo-parallel sentence pairs from monolingual corpora and learn translation without parallel data.

Step 1: Generate a Translation ENCODER

INPUT: **Clean Sentence_En**, Ex: I love pizza
OUTPUT: **Hidden States**_{cl-sent}



DECODER

INPUT: <IT> <START>
OUTPUT: Predict **Synthetic Translation_It**, Ex: Io Preferisco pizza

Step 2: Train Reverse Direction ENCODER

INPUT: **Synthetic Translation_It**, Ex: Io Preferisco pizza
OUTPUT: **Hidden States**_{syn-tr}



DECODER

INPUT: **Clean Sentence_En**: <START> I love pizza
(Teacher Forcing)
OUTPUT: **Predict Sentence_En**, Ex: I like pizza

Step 3: Back-Translation Loss

LOSS_BT-En: Prediction vs Original Clean Sentence : I like pizza vs I love pizza

Step 4: Repeat in Opposite Direction: Italian → English

$LOSS_{total} = L_{LM_En} + L_{LM_It} + L_{BT_En} + L_{BT_It} \Rightarrow$ **Update:** Encoder / Decoder / Embedding Parameters

3. The Two Models: 3.2 Phrase-Based Statistical Machine Translation

1. Initialization



Goal: Build an **Initial Phrase Table** for English $\leftrightarrow$ Italian **without parallel sentences**

Step 1: Tokenization English/Italian

Step 2: Train Monolingual Embedding Vectors: En/It corpus $\rightarrow$ [En/It Embeddings Vector]

Step 3: Align Embedding Spaces: **W:** $W(\text{Embedding_En}) \approx \text{Embedding_It}$; $W(\text{love}) \approx \text{amare}$

Nearest-Neighbor Retrieval: Cosine similarities: $\text{Cos}(W(\text{love}), \text{amare}) \dots (\text{IR})$

love $\rightarrow$ amare 0.92

love $\rightarrow$ piacere 0.71

love $\rightarrow$ adorare 0.15

....

love $\rightarrow$ casa 0.01

....

Step 4: Build Initial Bilingual Dictionary



Convert Similarities into Probabilities: $\sum P(\text{It}_w | \text{love}) = 1$ and Top-k selection



Initial Phrase Table

Initial Phrase Table

English	Italian	Prob
love	amare	0.55
love	piacere	0.42
love	adorare	0.03
good morning	buongiorno	0.75
good morning	buona mattina	0.25

3. The Two Models: 3.2 PBSMT

2. Language Modeling



Goal: Learn **fluent** Language (It/En) sentences from **monolingual corpora** and **Score** translation candidates

Step 1: Train a Statistical n-gram Language Model for each language

Stores n-gram statistics: how often words follow each other

$P(\text{buongiorno}) \times P(\text{ragazzi} | \text{buongiorno}) > P(\text{mattina}) \times P(\text{ragazzi} | \text{mattina})$

Step 2: Decoder Generates Translation Candidates from Phrase Table

good morning guys → buona mattina ragazzi, buongiorno ragazzi, ragazzi buongiorno

Step 3: Score Candidate Translations

Language Model → Candidate Fluency_Scores

(as it has learned **common word sequences** from corpora)

Reordering Model → Word Order_Scores

Step 4: Decoder (search Algorithm) Selects The Best Translation

Decoder Score = Log(Phrase Translation Score) + Log(LM_Fluency Score) + Log(W_Order_Score)

Decoder Winner: good morning guys → **buongiorno ragazzi**

Candidate_Trans	LM_Fluency_Score
buona mattina ragazzi	Low
buongiorno ragazzi	Higher

Candidate_Trans	W_Order_Score
ragazzi buongiorno	Low
buongiorno ragazzi	Higher

3. The Two Models: 3.2 PBSMT

3. Back-Translation



Goal: Improve translation quality by creating synthetic parallel corpora and updating the Phrase Table

Step 1: Build Synthetic Parallel Corpora

(Using the Level 2 PBSMT Translator)

1.1 Translate EN corpus → IT and create Synthetic Corpus A

1.2 Translate IT corpus → EN and create Synthetic Corpus B

(Called **synthetic parallel corpora** because one side is original and the other side is machine-generated)

Step 2: Re-Estimate the Phrase Table → Update phrase table

(So the phrase table becomes better)

Step 3: Build Better Translators

(Decoder translates using the updated Phrase Table)

Step 4: Repeat Iteratively step 1 → 3

Synthetic Corpus A

English	Synthetic Italian
good morning guys	buongiorno ragazzi

Synthetic Corpus B

Italian	Synthetic English
buongiorno tutti	good morning all

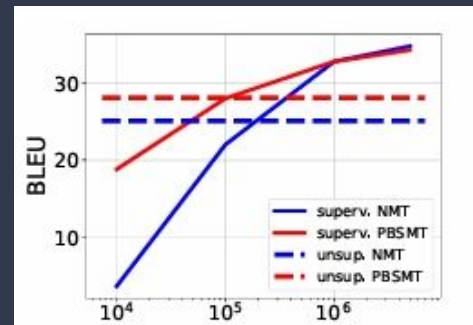
Updated Phrase Table

English	Italian	Prob
good morning	buongiorno	0.75 → 0.95↑
good morning	buona mattina	0.25 → 0.05↓

4. The Results and Comparison

4.1 Paper Results

4.2 Our Implementation Results



4.1 Paper Results: Table 2

1st Result: Proposed Unsupervised MT **Vs Previous Unsupervised MT**

Paper model	*BLEU for en-fr
Paper_NMT	25.1 vs 17.0
Paper_PBSMT	28.1 vs 17.0
Paper_Hybrid	27.6 vs 17.0

NMT

All paper models **outperform** previous unsupervised NMT.

Model	en-fr	fr-en	en-de	de-en
(Artetxe et al., 2018)	15.1	15.6	-	-
(Lample et al., 2018)	15.0	14.3	9.6	13.3
(Yang et al., 2018)	17.0	15.6	10.9	14.6
NMT (LSTM)	24.5	23.7	14.7	19.6
NMT (Transformer)	25.1	24.2	17.2	21.0
PBSMT (Iter. 0)	16.2	17.5	11.0	15.6
PBSMT (Iter. n)	28.1	27.2	17.9	22.9
NMT + PBSMT	27.1	26.3	17.5	22.1
PBSMT + NMT	27.6	27.7	20.2	25.2

2nd Result: Proposed NMT **Vs PBSMT** **Vs HYBRID Model**

Table 2: Comparison with previous approaches

1. **NMT**: Strong neural baseline
2. **PBSMT**: Surprisingly **stronger than NMT**!
3. **PBSMT + NMT**: The **Best Overall** big-picture model, **Especially for harder pairs**.

***BLEU**: is an **automatic MT evaluation score** that measures how similar a machine translation is to a human translation, by comparing n-gram overlap between machine and human translation.

4.1 Paper Results: Table 3

1st Result: The effect of the number of Back-Translation Iterations in PBSMT

Table 3 shows that **PBSMT improves** as the **number of iterations increases**

2nd Result: Final PBSMT Vs NMT Vs HYBRID Model

NMT: strong, but weaker for harder pairs

PBSMT: stronger and more stable than NMT

***PBSMT + NMT: Best overall** big-picture model (Especially for harder pairs.)

	en → fr	fr → en	en → de	de → en	en → ro	ro → en	en → ru	ru → en
<i>Unsupervised PBSMT</i>								
Unsupervised phrase table	-	17.50	-	15.63	-	14.10	-	8.08
Back-translation - Iter. 1	24.79	26.16	15.92	22.43	18.21	21.49	11.04	15.16
Back-translation - Iter. 2	27.32	26.80	17.65	22.85	20.61	22.52	12.87	16.42
Back-translation - Iter. 3	27.77	26.93	17.94	22.87	21.18	22.99	13.13	16.52
Back-translation - Iter. 4	27.84	27.20	17.77	22.68	21.33	23.01	13.37	16.62
Back-translation - Iter. 5	28.11	27.16	-	-	-	-	-	-
<i>Unsupervised NMT</i>								
LSTM	24.48	23.74	14.71	19.60	-	-	-	-
Transformer	25.14	24.18	17.16	21.00	21.18	19.44	7.98	9.09
<i>Phrase-based + Neural network</i>								
NMT + PBSMT	27.12	26.29	17.52	22.06	21.95	23.73	10.14	12.62
PBSMT + NMT	27.60	27.68	20.23	25.19	25.13	23.90	13.76	16.62

Table 3: Fully unsupervised results

Table 3 confirms the results of Table 2 for more dataset, that **combining PBSMT with NMT**(Not inverse) **often gives** the **strongest overall unsupervised MT system**

***PBSMT + NMT:** add the **data generated** by the unsupervised **PBSMT** system to the **back-translated** data produced by the **NMT** model

4.1 Paper Results: figure 2

Unsupervised Models Competes with Supervised Models

- When **parallel data is small (low-resource languages)**, the **unsupervised models** can **outperform** supervised models.
- **Unsupervised PBSMT** achieves **similar performance** to **supervised PBSMT** trained with more than **100,000 parallel sentences**.
- But when the **supervised model has a large parallel corpus**, especially around **1000,000** sentence pairs, supervised MT **becomes stronger**.

Examples:

Case	Parallel Data Used By Supervised	Supervised BLEU	UnSupervised BLEU	
Romanian → English	6,000	22.9	23.9	Outperforming UnSupervised method by 1 point!
Urdu → English	800,000 noisy data	9.8	12.3	both low resource and distant

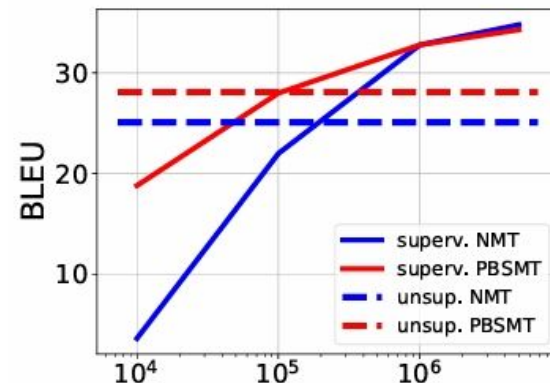


Figure 2: Comparison between **supervised** and **unsupervised** approaches on WMT'14 En-Fr, as we vary the **number of parallel sentences** for the **supervised methods**.

Our Implementation Results: Toy NMT

Goal: Implement a **light version** of the paper's NMT idea on a small **English–Italian** dataset.

Model Setup:

1. **Small supervised warm-up:** learns a basic EN↔IT translation mapping with one shared embedding layer for both languages.
2. **Denoising autoencoder:** learns fluency from noisy monolingual sentences
3. **Light back-translation:** creates synthetic sentence pairs and retrains the model

Evaluation:

Number of Sentences	BLEU
50	0.47
100	0.58
250	0.63
500	0.84

Our Implementation Results: Toy PBSMT

Goal: Implement a **light version** of the paper's PBSMT idea on a small **English–Italian** dataset.

Model Setup:

1. **Initialization:** builds an **initial EN↔IT phrase table** using aligned Word2Vec embeddings and **nearest-neighbor retrieval**.
2. **Language modeling + decoding:** uses a **Bigram language** model to score fluent Italian translation candidates.
3. **Bidirectional back-translation:** creates **synthetic EN→IT and IT→EN pairs** and updates the phrase table.

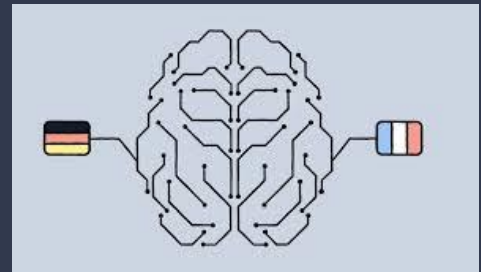
Evaluation:

Number of Sentences	BLEU
50	1.34
100	2.00
250	1.68
500	1.52

Result:

- **Toy_PBSMT** gives **higher BLEU** than the **Toy_NMT**.
- **Back-translation** does not always improve the result because the **synthetic translations** are **still noisy**.

5. Conclusion



5. Conclusion

The contribution of this **paper** is that it **combines** and **improves 3 principles** in a unified framework for **Unsupervised NMT** and **PBSMT**.

- **Initialization** (shared lingual embeddings/initial phrase table)
- **Language modeling**
- **Iterative bidirectional back-translation**

Result 1: The **proposed unsupervised MT** models strongly **outperform** previous **unsupervised MT** methods.

Result 2: **PBSMT** is surprisingly **stronger** than **NMT** in this setting, while **PBSMT + NMT** gives the **best** overall big-picture performance (Especially for harder pairs).

Result 3: Table 3 confirms the results of Table 2 for more dataset, that **combining PBSMT with NMT**(Not inverse) **often gives the strongest overall unsupervised MT** system

Result 4: When **parallel data is small** (**low-resource languages**), the **unsupervised models** can **outperform** supervised models, like for **Romanian → English** or **Urdu → English**

Result 5: Our **Toy PBSMT** achieves **higher BLEU** than our **Toy NMT**, which is **consistent** with the **paper's finding** that **PBSMT** can **outperform NMT** in low-resource unsupervised settings.

Reference

Phrase-Based & Neural Unsupervised Machine Translation

Guillaume Lample, Myle Ott, Alexis Conneau, Ludovic Denoyer, Marc'Aurelio Ranzato

arXiv:1804.07755v2

Thank you so much for your attention

